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Simscape Model of System Control of Underwater Oceanic Habitat

Detailed Design of Control System

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1 Technical Summary

Table 1: Reflection questions on the detailed design simulation.

What part of the facility is simulated by your detailed design?	The model simulates the control system behind the underwater habitat, which includes the communication system, docking system and control (SCADA) system.
What are the key features of your simulation?	The simulation includes a comprehensive logic distribution system where three branches act as configurations for the three main modes of the system (Normal Operations, Docking System and Emergency Mode)
How realistic is your simulation, and how could it be made more realistic?	The simulation is built to take into account some of the real life effects that might affect the model, for example all loads have a thermal mass attached to the axis as the temperature effect. In addition the loads are modelled as variable resistors to act as fluctuations to simulate real life. To make the simulation more realistic, some more real life scenarios can be taken into account, like habitat temperature affects load's resistance, which affects the amount of power each draws. Simulating the docking location and approach to automatically trigger the docking sequence instead of manual input would have made the control logic more useful.
What did you find were the best approaches to modelling to improve power efficiency?	The most efficient approach was to select and use variable resistor loads with values that reflected realistic power draws for each subsystem and ensuring that the converter output is stable across all three operating modes. Doing this minimized resistive losses and kept the efficiency above 92% for all operating conditions
What obstacles did you encounter and spend the longest time getting to work?	The longest aspect of the model was setting up the controller system, building the blocks, ensuring that it affected all the right branches, and validating all the results. Furthermore the docking system wasn't implemented as originally conceptualised due to the process of adding different modes and triggers failing, and therefore simplified to only branch into two components (electromagnetic clamps and charging inductor).
How would you tackle the simulation differently if you could start again?	Given the opportunity to restart, the control logic of the circuit would be implemented first to ensure sound command over all three branches. Moreover, modelling the docking subsystem in a simple way first to make sure the loads, voltage and power going towards the branches all made sense before setting up the control logic would have benefited in reducing the amount of time used up on simulating the circuit.

2 Simulation

2.1 Quantitative Description: Specification Table

The simulation models the power distribution across the loads presents in the Underwater Oceanic Habitat's Control System. Table 2 presents the specification of the simulation, detailing the quantitative values and system configuration across all three operating modes.

Table 2: Specification Table for Systems Simulation.

Maximum Parameter	Normal	Dock	Emergency
Total Power Used by Loads [W]	125.38	322.5	73.68
Power Input to Converter [W]	145.77	470.9	82.9
Power Output by Converter [W]	128.5	347.35	75.7
Voltage Input to Converter [V]	48.97	48.9	48.9
Voltage Output by Converter [V]	23.6	22.48	23.7
Current Input to Converter [A]	3.03	9.76	1.76
Current Output to Converter [A]	5.48	15.47	3.2
Branch 1 Voltage [V]	23.12	22.78	23.24
Branch 1 Current [A]	3.167	3.12	3.2
Branch 2 Voltage [V]	23.29	23.05	0
Branch 2 Current [A]	2.589	2.48	0
Branch 3 Voltage [V]	0	22.5	0
Branch 3 Current [A]	0	10.23	0
Docking Subsystem Power [W]	0	207.8	0
Communication System Power [W]	45.79	41.89	0
Data Logger Power [W]	13.7	12.6	0
Sensors Power [W]	18.4	22.38	23.27
Controller Power [W]	31.6	30.7	32.25
Protection Monitoring Power [W]	18.56	18.02	18.7
Charging Power [W]	0	143.3	0
Clamps Power [W]	0	79.4	0
Current Mode [1,2,3]	1	2	3
Fault [0,1]	0	0	1
Location of Fault [0,1,2,3]	0	0	2
Efficiency [%]	97.6	92.8	97.3

Category	Description
Subsystems	Control System
	Communication System
	Docking System
	Sensor Management
Load Distribution	Branch 1: Critical Operations (always active)
	Branch 2: Communication System (isolated in Emergency)
	Branch 3: Docking Procedure (active in Dock mode only)
Control Signals	Mode control signal [1, 2, 3]
	Fault signal [0, 1]
	Location of Fault [0, 1, 2, 3]
	Closed-loop mode switching based on fault feedback
Protection	Automatic branch isolation on fault detection
	Branch 2 fully isolated under fault condition
	Protection monitoring active across all modes
Thermal Mass	Protection Monitoring [200 J/K]
	Controller + I/O [270 J/K]
	Sensors [90 J/K]
	Data Logger [180 J/K]
	Communication System [500 J/K]
	Charger [900 J/K]
Voltage Conversion	$\approx 2:1$ step-down across all modes

2.2 Model: The Simulation model diagram(s)

2.2.1 Overall Model

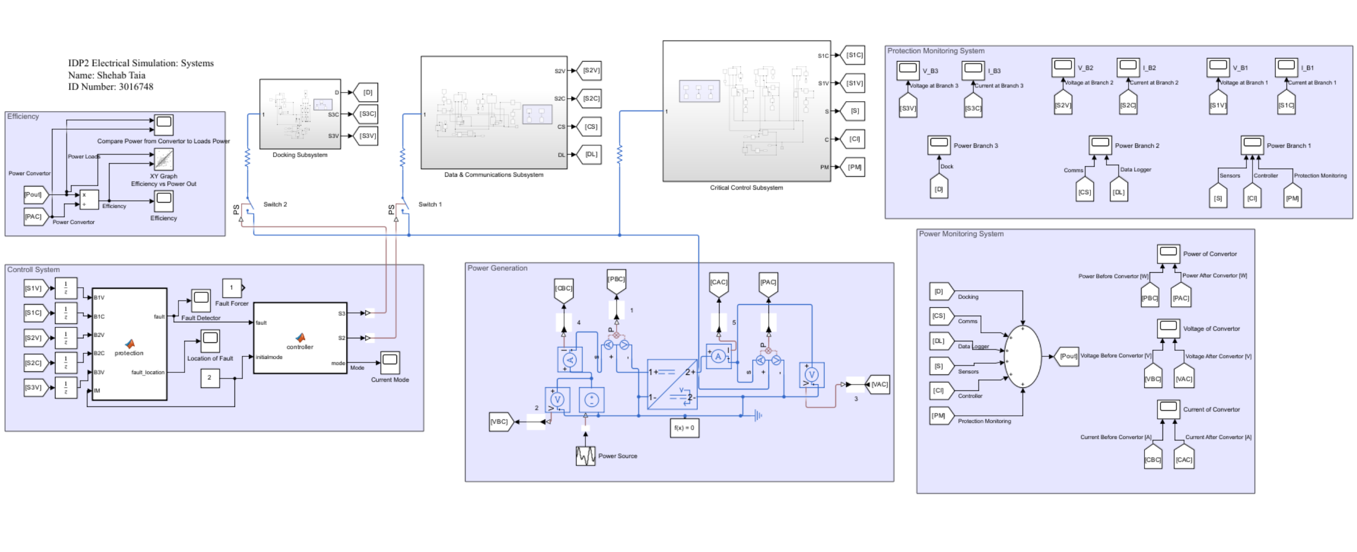

Figure 1: Overall Simulation for System

The overall model simulates the power distribution system of the control logic of the SCADA, communications, docking and sensor management of the underwater habitat. The system is powered by a 48V sinusoidal source fed into a DC-DC converter that steps the voltage down to 24V. This is then supplied to three parallel load branches, where a control system monitors all branches' voltages and currents to detect any faults and appropriately handle them. The system operates on three different modes: Normal operations (Mode 1), Docking Procedure (Mode 2) and Emergency Mode (Mode 3).

2.2.2 Power Generation

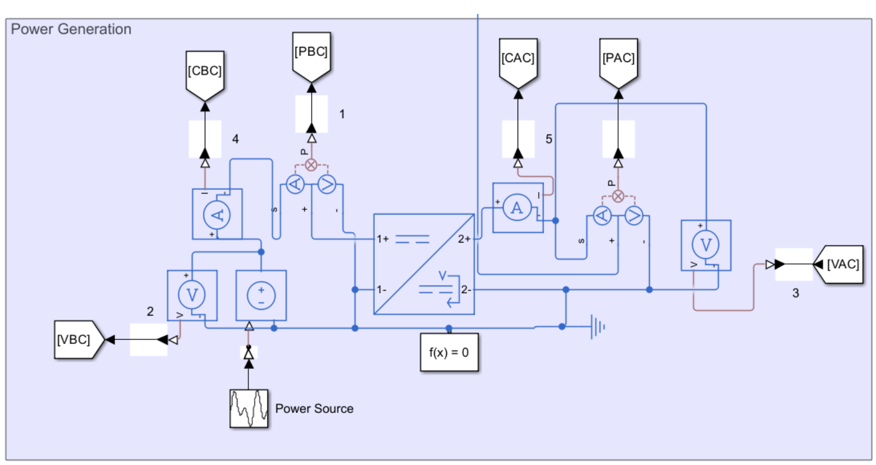

Figure 2: Power Generation for System

The power is supplied from the Energy subsystem (WP1) as a 48V DC source that is modelled as a controlled voltage source that is powered by a wave generator with expression $48 + \sin(1, 0.5, 0)$. This initiates a small oscillation to represent real world fluctuations inherent in power generation rather than using a simple idealised flat DC source. Three sensors are used: voltage, current, and power sensors shown as (VBC, CBC, PBC), respectively. These measure the source output before it is fed into the converter.

2.2.3 Power Conversion

The DC DC converter block is modelled in the same power generator subsystem as the voltage source and can be seen in Figure 2. The block steps the 48V down to 24V with droop control enabled to introduce realistic load dependent voltage variations which are seen in the different results gathered in all three modes. The converter is also configured to have an efficiency of 80% which reflects realistic power losses between input and output. The output of the converter is then fed to the three different load subsystems. Protection monitoring is used to measure the outputs of the converter using the sensors (VAC,PAC,CAC).

2.2.4 Load

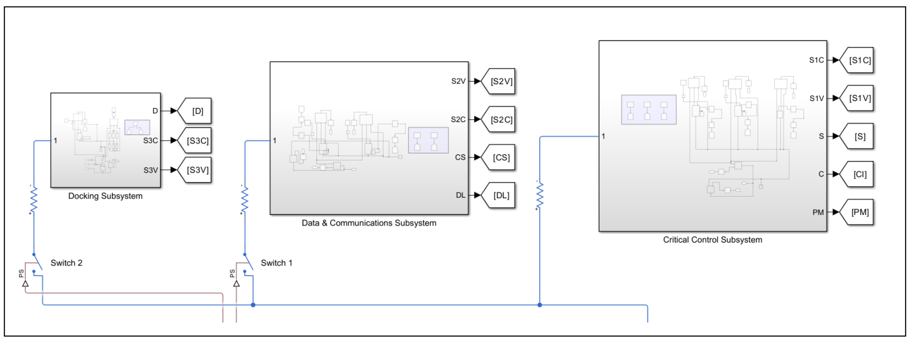

Figure 3: All Loads Subsystems

The load system controls three functional groups that are modelled as three parallel branches. Connecting them in parallel allows for individual switch control without affecting any other branch. The majority of loads have been modelled using a variable resistor with fluctuations with the addition of thermal mass to simulate realistic working conditions.

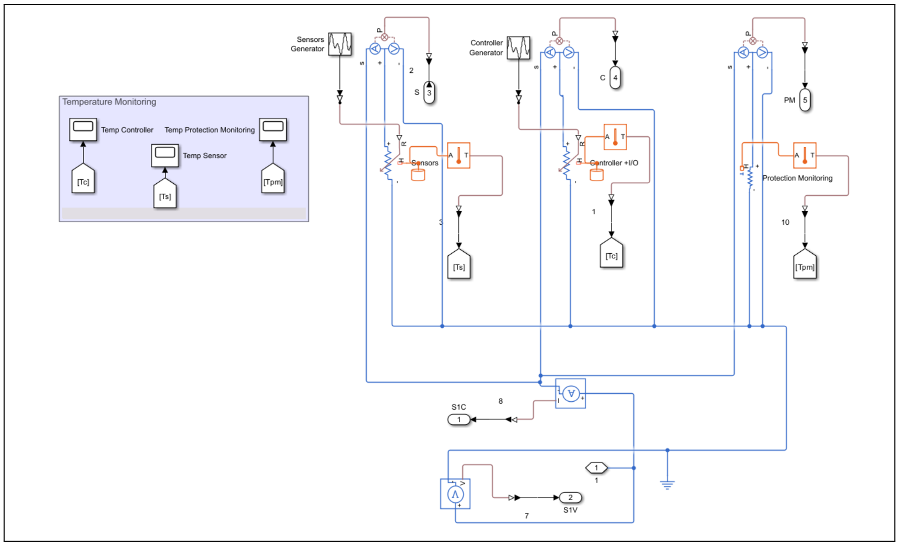

Figure 4: Critical Loads Subsystem

Branch 1 consists of the critical systems: SCADA controller ($\approx$ 30W), Sensors ($\approx$ 20W) and protection monitoring system ($\approx$ 18W). These components are essential for maintaining habitat operation and safety therefore they must remain on at all times.

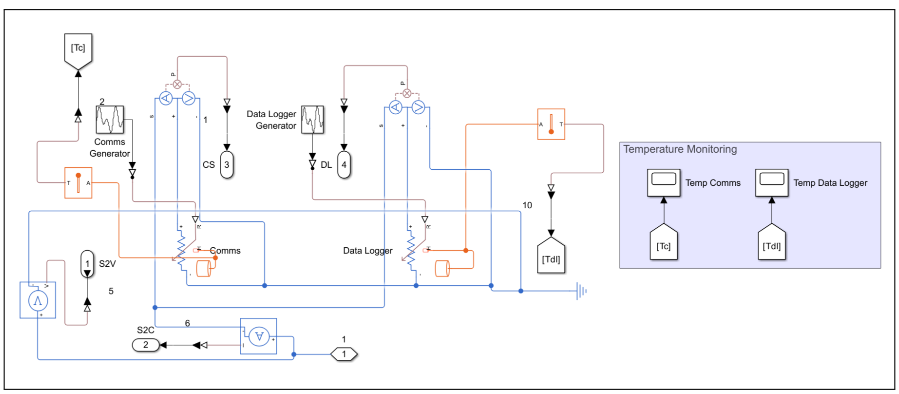

Figure 5: Communication and Data Subsystem

Branch 2 contains the communication and data logger subsystem. The communication variable resistor is modelled using a wave generator to represent the communication networks interchanging between standby and burst modes. This branch is always active except when the controller triggers a fault; the branch is fully isolated as the subsystem is considered non-essential therefore it is shed to prioritise power to the critical system. The maximum power drawn from communication is ($\approx$ 45W) and the data logger draws ($\approx$ 13W).

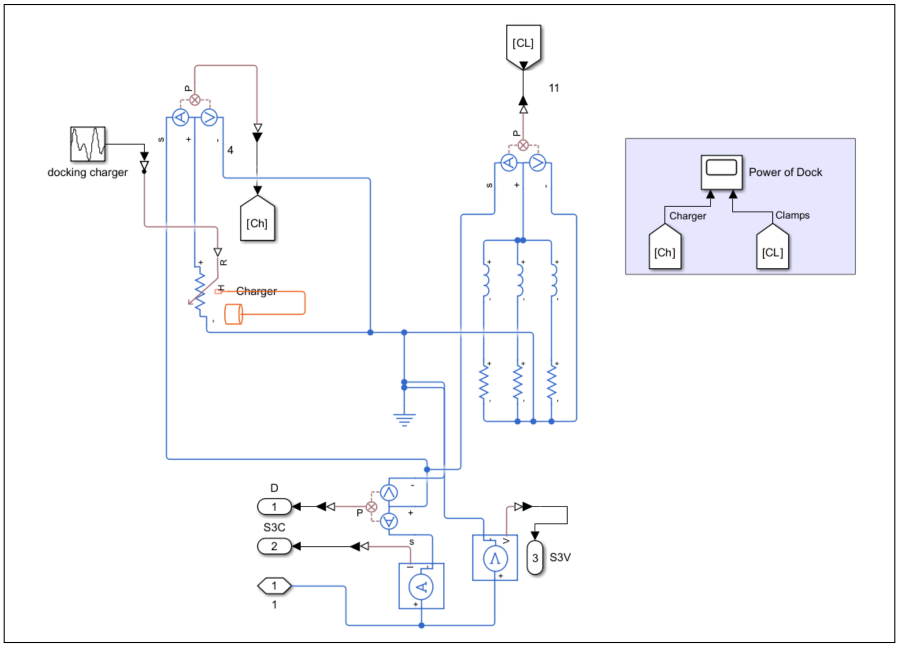

Figure 6: Docking Subsystem

Branch 3 represents the docking system which only activates during Docking mode (Mode 2). It is made up of three electromagnetic RL circuits (≈ 80W) which are used to secure ROVs in place during the docking procedure. And an inductive charging station (≈ 145W) which is being modelled as a variable resistor to simulate the charging duration of an ROV.

2.2.5 Control

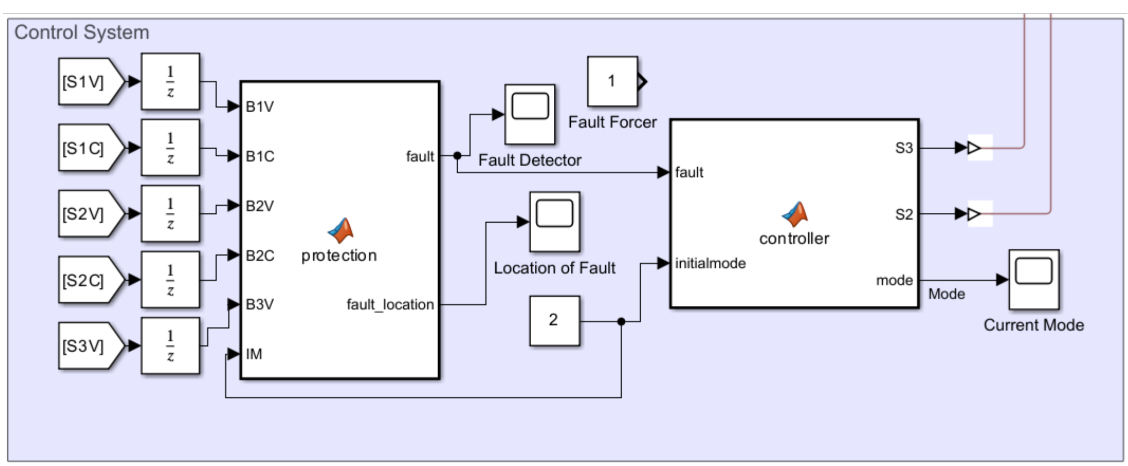

Figure 7: Controller

The control system uses two MATLAB function blocks to implement a closed loop feedback control. Firstly branch voltages and currents (S1V, S1C, S2V, S2C, S3V) which can be seen in Figure 8 are fed through a unit delay block before passing into the protection block, which evaluates whether each value is within a predetermined acceptable range. If any value exceeds this range the block outputs a fault signal and identifies which branch contains the fault via fault_location. These signals are then passed to the controller block which determines what mode should be running and controls the branch switches accordingly. The current mode is then displayed via a scope. A Fault Forcer constant block can be used to manually inject a fault signal for testing.

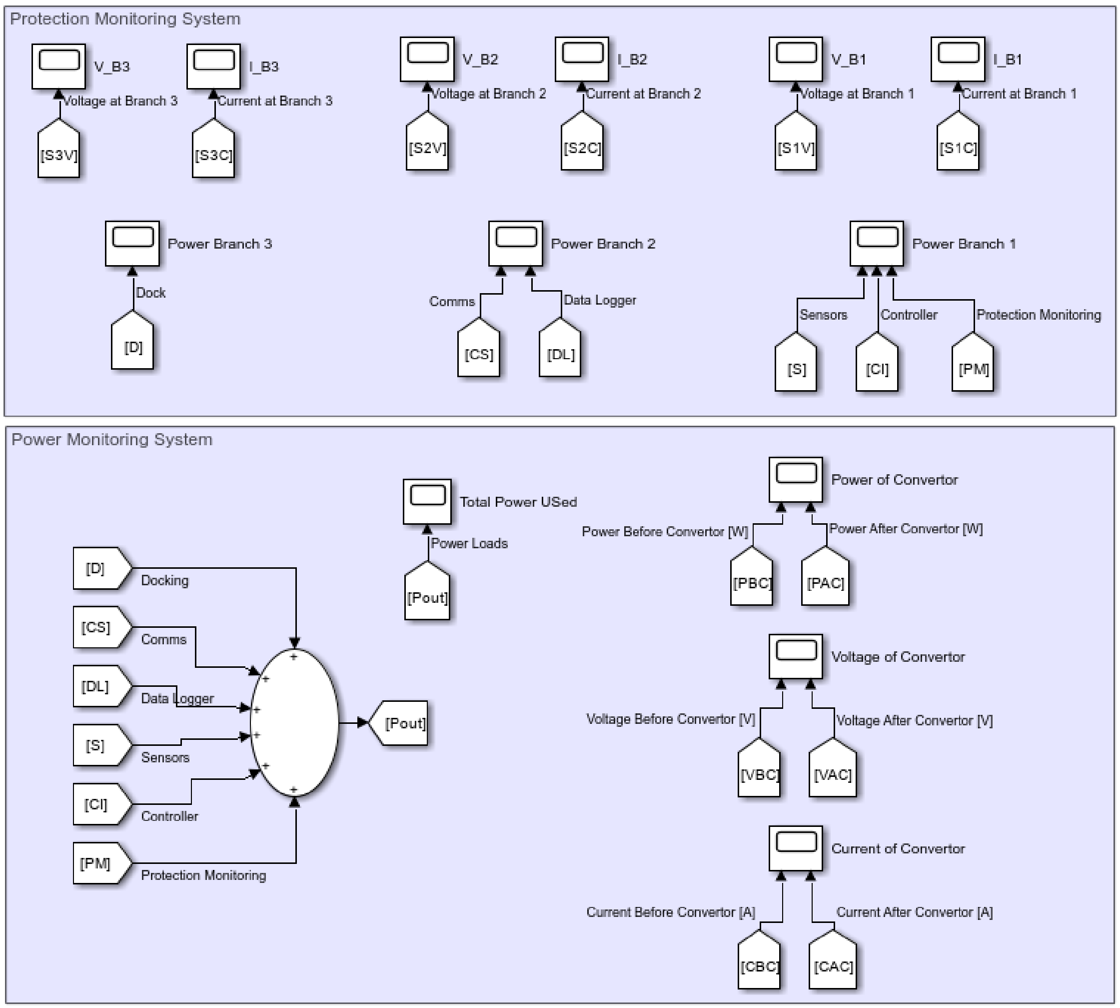

Figure 8: Protection Monitoring System

2.3 Analysis: The Simulation Result

2.3.1 Inputs and Outputs

Figures 9–11 display the total load power (Pout) across all three operating modes. The fluctuations seen are a reflection of a combination of trigger modes (communication turning on) and realistic variable resistor wave generators. The steady state power achieved is 125.38W, 322.5W, and 73.68W in Normal, Dock, and Emergency modes respectively, confirming the correct branch switching behaviour. When branch 3 is active the power significantly increases in dock mode. In mode 3 both branches 2 and 3 are isolated, so only the critical system remain powered which demonstrates the effectiveness of the fault-triggered load shedding control logic.

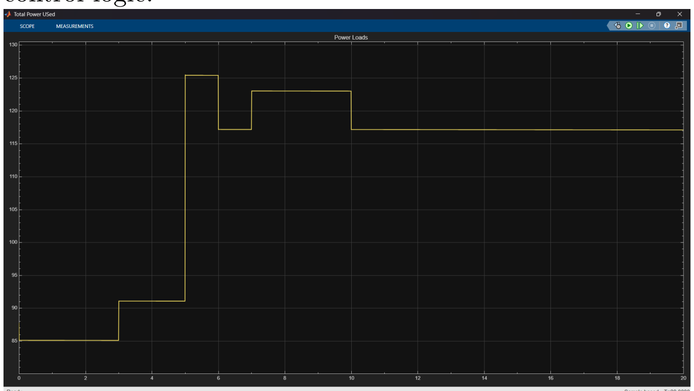

Figure 9: Pout Mode 1

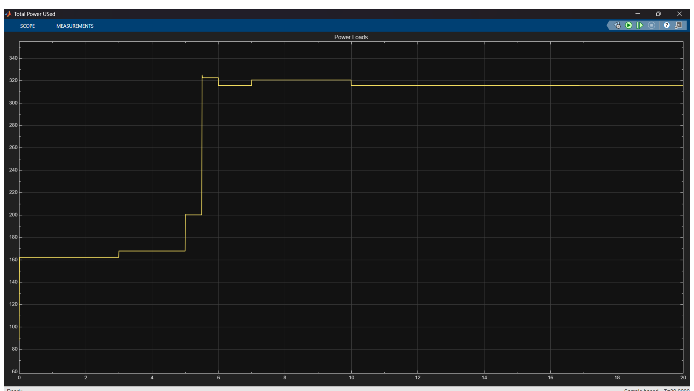

Figure 10: Pout Mode 2

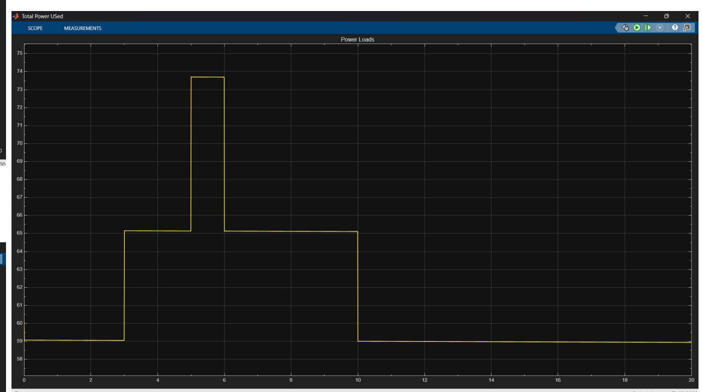

Figure 11: Pout Mode 3

2.3.2 Energy Efficiency

Figures 12–14 show the efficiency (Pout/PAC) over time for each mode while Figures 15–17 show the XY efficiency against power graphs. All three modes show a clear negative relation between efficiency and load power which is caused by droop control introducing a larger voltage drop when under higher currents. In Normal and Emergency modes the efficiency is ≥0.976 and ≥0.973 respectively due to lower load demands. However when Dock mode is active the efficiency drops to a minimum of 0.928 which is driven by Branch 3 activation. The closed loop fault detection system therefore improves the efficiency by isolating non essential branches directly minimising unnecessary power losses during Emergency mode.

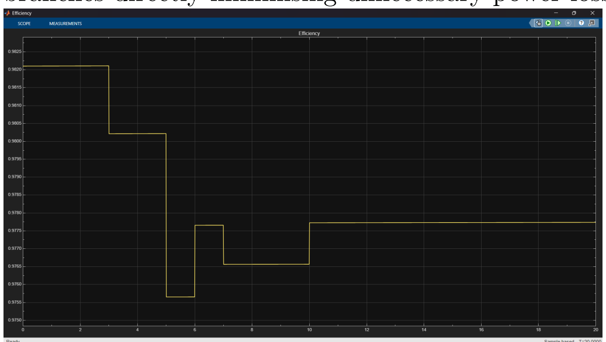

Figure 12: Efficiency Mode 1

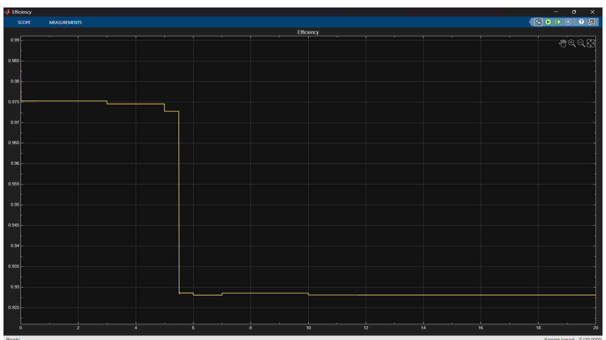

Figure 13: Efficiency Mode 2

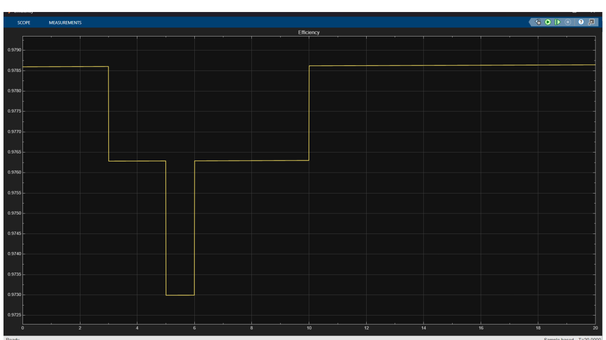

Figure 14: Efficiency Mode 3

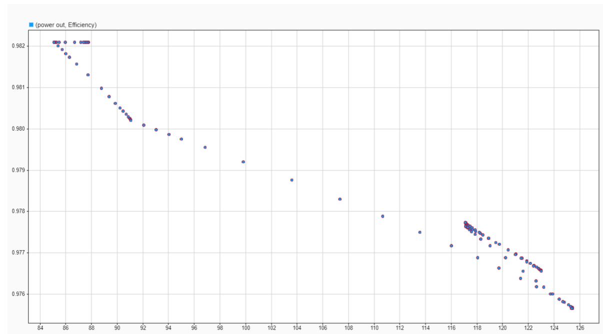

Figure 15: XY Mode 1

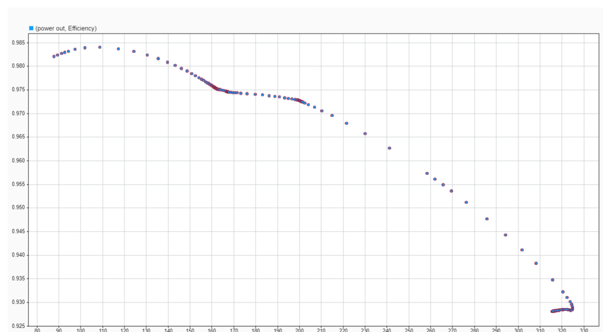

Figure 16: XY Mode 2

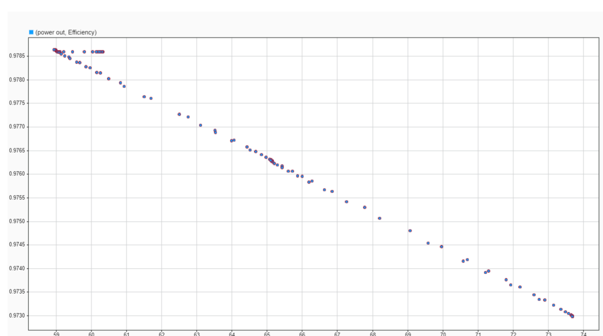

Figure 17: XY Mode 3

A Flowchart of Overall System

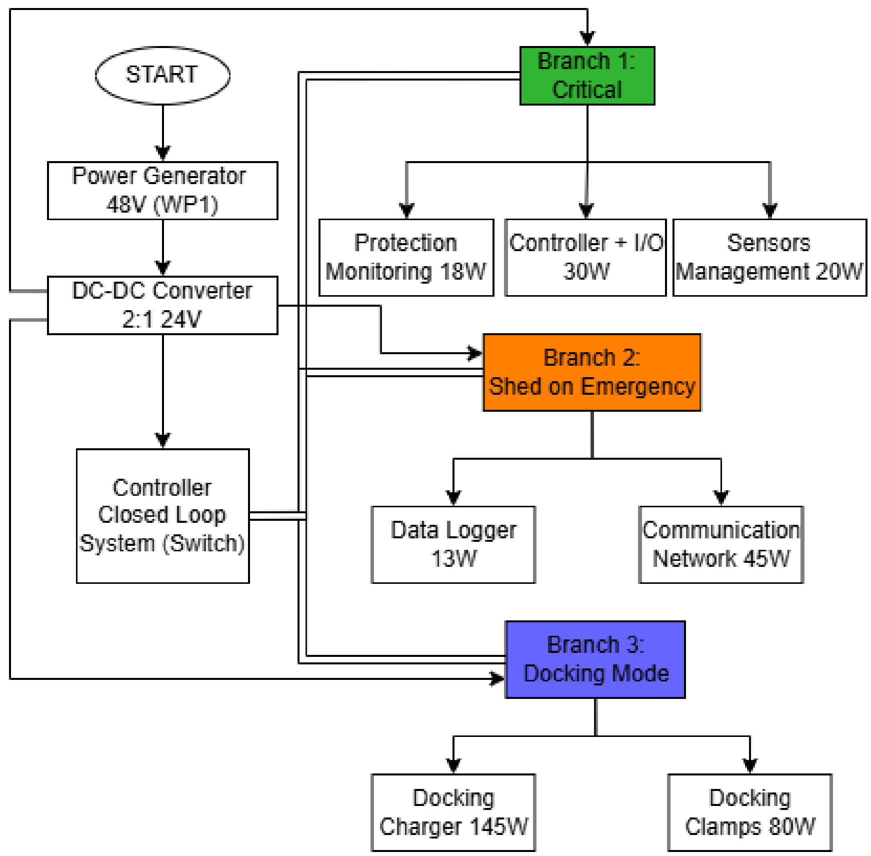

Figure 18: Flowchart of Power Distribution and Control of System

B MATLAB Controller Code

```
function [fault, fault_location] = protection(B1V, B1C, B2V, B2C, B3V, IM)
fault = double(0);
fault_location = double(0);
if B1V < 21 || B1C < 2
    fault = double(1);
    fault_location = double(1);
elseif B2V < 21 || B2C < 1
    fault = double(1);
    fault_location = double(2);
elseif IM == 2 && (B3V < 20)
    fault = double(1);
    fault_location = double(3);
else
    fault = double(0);
    fault_location = double(0);
end
```

Listing 1: Protection MATLAB Function

```
function [S3, S2, mode] = controller(fault, initialmode)
S2 = double(1);
S3 = double(0);
mode = double(1);
if fault == 1
    mode = double(3);
    S2 = double(0);
    S3 = double(0);
elseif initialmode == 2
    mode = double(2);
    S2 = double(1);
    S3 = double(1);
else
    mode = double(1);
    S2 = double(1);
    S3 = double(0);
end
```

Listing 2: Controller MATLAB Function